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Research

Strong extrinsic reproductive isolation between parapatric populations of an Australian groundsel

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Summary

- Speciation with gene flow, or the evolution of reproductive isolation between interbreeding populations, remains a controversial problem in evolution. This is because gene flow erodes the adaptive differences that selection creates between populations.
- Here, we use a combination of common garden experiments in the field and in the glasshouse to investigate what ecological and genetic mechanisms prevent gene flow and maintain morphological and genetic differentiation between coastal parapatric populations of the Australian groundsel *Senecio lautus*.
- We discovered that in each habitat extrinsic reproductive barriers prevented gene flow, whereas intrinsic barriers in F_1 hybrids were weak. In the field, herbivores played a major role in preventing gene flow, but glasshouse experiments demonstrated that soil type also created variable selective pressures both locally and on a greater geographic scale.
- Our experimental results demonstrate that interfertile plant populations adapting to contrasting environments may diverge as a consequence of concurrent natural selection acting against migrants and hybrids through multiple mechanisms. These results provide novel insights into the consequences of local adaptation in the origin of strong barriers to gene flow in plants, and suggest that herbivory may play an important role in the early stages of plant speciation.

Introduction

Local adaptation is thought to be a major contributor to the evolution of reproductive isolation (RI) between parapatric populations (Rundle & Nosil, 2005; Schemske, 2010). Although gene flow usually turns the odds against the formation of new species (Felsenstein, 1981), ecological contributions to RI in parapatry are not uncommon, having a long history in natural and experimental settings. For instance, adjacent populations of grasses on soils containing high or low concentrations of toxic heavy metals have evolved into morphologically and physiologically differentiated populations that persist despite gene flow (McNeilly & Antonovics, 1968; Antonovics & Bradshaw, 1970; Antonovics, 2006). Similarly, *Anthoxanthum odoratum* populations exposed to different environmental conditions have evolved both morphological differences and RI since the inception of the Park Grass Experiment in 1856 (Davies & Snaydon, 1976; Silvertown *et al.*, 2005). These empirical results echo those from theory, where the predicted conditions for parapatric speciation seem to be common in nature (e.g. isolation by distance between populations, and patchy and linear habitats along rivers and coasts; Gavrilets *et al.*, 2000). However, studies of RI often fail to identify the agents of divergent natural selection or quantify the relative contributions of multiple reproductive barriers to gene flow during the lifetime of organisms (c.f. studies reviewed in Lowry *et al.*, 2008a). As a consequence, our knowledge of which barriers trigger the speciation process, and the relative importance of extrinsic vs intrinsic barriers to gene flow remains limited in most studies of ecotype and species formation in plants.

Local adaptation creates barriers to gene flow in parapatry through various mechanisms (Schluter, 2001; Rundle & Nosil, 2005; Hendry *et al.*, 2007). When locally adapted populations exchange migrants, theory predicts that they will fare poorly in the environment of the sister population. This creates greater opportunities for interbreeding within rather than between populations, in turn limiting gene flow (Nagy & Rice, 1997; Hendry, 2004; Nosil, 2004; Thibert-Plante & Hendry, 2009). Local adaptation can also cause ecologically dependent reductions in F_1 hybrid fitness, a phenomenon known as extrinsic postzygotic RI (Schluter, 2000; Rundle & Whitlock, 2001; Rundle & Nosil, 2005). Generally, ecologically dependent reductions in hybrid fitness occur because hybrids express intermediate parental phenotypic values for locally adapted traits (Barton & Hewitt, 1985; Schluter, 2000; Rundle & Nosil, 2005), thus rendering them unfit in parental habitats. The extent to which hybrids are ecologically disadvantaged depends on the form of inheritance for the traits under divergent natural selection (e.g. dominance vs additivity; Arnold, 1997; Barton, 2001; Berner *et al.*, 2011). However, reductions in F_1 hybrid fitness could also result from their failure to cope with stressful conditions, including those

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experienced in the field (Coyne & Orr, 2004). This form of hybrid failure is also extrinsic but not specific to the environment of the parents that produced the hybrid offspring. These mechanisms of postzygotic RI only manifest under field or stressful conditions but dissipate under controlled or benign conditions, such as those found in glasshouses (Hoffmann & Merilä, 1999; Bordenstein & Drapeau, 2001).

The mechanisms creating divergent natural selection are often difficult to identify and quantify. However, some habitat differences are known to contribute to local adaptation in plants (Kruckeberg, 1986; O'Dell & Rajakaruna, 2011). Edaphic and climatic differences usually cause strong divergent selection between populations, possibly because of drought (Stebbins, 1952; Bray, 2002), toxicity (Brady *et al.*, 2005), temperature (Keller & Seehausen, 2012) or a combination of them. These effects are common across many plant taxa (Kruckeberg, 1951; Wu *et al.*, 1975; Emms & Arnold, 1997; Nagy & Rice, 1997; Vekemans & Lefèbvre, 1997; Berglund *et al.*, 2004; Kay, 2006; Martin *et al.*, 2006; Sambatti & Rice, 2007; Lowry *et al.*, 2008a) and suggest that environmental stress could create strong extrinsic prezygotic and varying degrees of postzygotic extrinsic RI, thus playing an important role during the early stages of ecotype and species formation (Lowry, 2012).

Plant and animal interactions can also contribute to the evolution of RI between plant populations. The most famous examples involve systems where pollinators discriminate floral differences between ecotypes or species (Emms & Arnold, 1997; Bradshaw & Schemske, 2003; Kephart & Theiss, 2004; DellíOlivo *et al.*, 2011). A less studied biotic cause of RI is the contributions of herbivores and parasites to reductions in gene flow between populations (Sork *et al.*, 1993; Combes, 1996; Fritz *et al.*, 1999; Elias *et al.*, 2012). For instance, hybrids between populations of willows show differential responses to aphids and mites (Fritz *et al.*, 1994; Czesak *et al.*, 2004), similar to the adult hybrids of *Oenanthe conioides* and *O. aquatica* that are preferentially grazed by waterfowls and snails in the environment of *O. conioides* (Westberg *et al.*, 2010). Further exploration is required to conclude whether this kind of interaction between plants and invertebrates is important for the progress towards speciation and the origins of RI in plants.

Populations of the groundsel *Senecio lautus* that inhabit the sandy dunes (Dune populations) and rocky headlands (Headland populations) along the Australian coast are an excellent system to study the origin and maintenance of ecotypes and the early stages of speciation. Often found adjacent to each other along the coast, Dune and Headland populations show marked morphological differentiation (Thompson, 2005; Supporting Information Fig. S1 for typical ecotype morphologies), which has evolved repeatedly and independently multiple times, and in the face of gene flow (Roda *et al.*, 2013a). Dune and Headland populations retain their morphologies in glasshouse conditions (see Abbott, 1976 for a similar case in European *Senecio*), and reportedly exhibit weak intrinsic reproductive barriers (Ali, 1964, 1968). Previous transplant experiments in *S. lautus* suggest that some populations are adapted to their local environment (Radford *et al.*, 2004). However, little is known about what causes differentiation in the system and whether local adaptation is driving the evolution of RI in parapatric populations. We chose one of these parapatric pairs to investigate the evolution of reproductive isolating barriers in response to adaptation to contrasting coastal environments. Through field observations and common garden experiments in the field and the glasshouse we estimated various components of RI between these two parapatric populations, and identified possible ecological mechanisms causing divergent natural selection. We discuss how these results inform us about the relative contributions of extrinsic and intrinsic reproductive barriers during the early stages of speciation with gene flow.

Materials and Methods

The system

Senecio lautus (G.Forst. ex Willd.) is a diverse complex of groundsels inhabiting a wide variety of environments in the South Pacific. Populations in Australia occupy diverse habitats including alpine and tablelands, woodlands, and coastal sand dunes and cliffs (Ali, 1964; Radford *et al.*, 2004; Thompson, 2005). Most populations exhibit strong self-incompatibility (Ornduff, 1964; and personal observations), and display variable life history traits ranging from annual and biennials to short- and long-lived perennials (Ali, 1968). Traits such as leaf morphology and plant architecture show strong associations with the environment in which populations are found, suggesting that the system consists of multiple ecotypes distributed across geography (Radford *et al.*, 2004; Roda *et al.*, 2013a). Populations inhabiting sand dunes (Dune ecotype) and rocky headlands (Headland ecotype) are related by geography and not habitat, and each pair is genetically differentiated from such other pairs (Roda *et al.*, 2013a). The pair at Cabarita Beach displays strong heterogeneous divergence across its genome and moderate average F_{st} (0.04) compared with comparisons between allopatric populations (F_{st} 0.12–0.2; Roda *et al.*, 2013a). We studied two coastal forms of *S. lautus* that grow on the sand dune (28°19′54.66″S 153°34′17.04″E) and rocky headland (28°21′45.07″S 153°34′46.82″E) environments at Cabarita Beach, in northern New South Wales, Australia (Fig. S1). Although the beach and headland are abutting environments, a small town separates the Dune and Headland populations by *c.* 800 m. The beach and parts of the headland remain connected via rocky outcrops and small cliffs. Of note, unpublished data suggest that these populations and similar pairs, are at least 30 000 yr old, whereas towns in Australia are < 200 yr old, suggesting that effects from urbanization on population dynamics are recent compared with those that drove past evolutionary divergence. Usually only a few metres separate other population pairs along the coast where towns do not interrupt the shore (e.g. the population pair at Coffs Harbour, NSW are separated by 3 m). The headland habitat is characterized by rocky mineral-rich soil, exposed to constant salt spray and strong winds. In this habitat individuals exhibit a compact architecture, being short and matt forming (prostrate), heavily branched, with small succulent leaves. In the dune habitat, the soil is sandy, poor in nutrients and susceptible to heating during sunny days. Individuals from

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the sand dunes are tall and have few branches with large and thin leaves (Radford *et al.*, 2004; Thompson, 2005; Fig. S1).

Crosses

Seeds were collected from 30 individuals from each coastal population of Cabarita Beach in 2009. Laboratory seed stocks for parental and F_1 hybrids full-sib families –derived from randomized crosses with equal contribution between parental types– were created in glasshouses at the University of Queensland, St. Lucia, QLD, Australia. Scarified seeds (1 mm trimmed at the micropyle side) from each family were germinated on moist filter paper in petri dishes. Seeds were kept in dark and controlled conditions for 3 d to induce root elongation, and subsequently placed under the light for 7 d to induce vegetative growth. One-week-old seedlings were transplanted into 2.5 l pots filled with standard potting mix and transferred to a glasshouse with constant temperature (25°C) and 12 h : 12 h, light : dark cycle. Flowering time was recorded, and plant morphology measurements were made on adult individuals with flowers (*c.* 2 months after germination). Intra- and interpopulation crosses were performed twice a day by gently rubbing flower heads (capitula): each flower head was crossed at least three times to maximize the number of florets producing seeds. We kept track of unpollinated flowers to check for self-pollination, but did not find a single seed resulting from self-pollination.

Intrinsic RI

We calculated seed set by estimating the proportion of fertilized seeds in flower heads (Fig. S2). We divided the number of fertilized seeds in an interpopulation cross by the average number of seeds produced in parental intrapopulation crosses (Coyne & Orr, 2004). We calculated F_1 seed set for two other parapatric population pairs (Hat head and Lennox Head) and two allopatric comparisons (Cabarita beach vs Byron Bay and Lamington National Park vs Port Macquarie) to provide context for our focal result (see Table S1 for the geographic location of all populations). The Tableland population from Lamington National Park, another member of the *S. lautus* species complex, is a perennial ecotype found far from the coast next to the edges of tropical forests. It can reach several metres in height, and in contrast to coastal types, has long and ovate serrated leaves (Radford *et al.*, 2004; Roda *et al.,* 2013a).

Flowering time differences between ecotypes

In order to investigate if the natural populations are separated phenologically, we counted the number of flowers per individual each month between February 2011 and January 2012. We report the total number of flowers per month in the population divided by the total number of individuals to control for changes in population size during the year. Because Cabarita Beach sand dunes are linear and narrow, we sampled flowers in individuals present in the same 80 m × 4 m transect. On the Headland population we sampled individuals in three different rocky grooves where the majority of individuals resided. We could not sample a few individuals growing on the cliffs due to high risk of falling.

Extrinsic RI

Soil experiments We performed reciprocal germination and establishment experiments under controlled conditions in the glasshouse using soil collected in each locality. We filled two sets of eight plastic trays with fresh soil from either the sand dunes or the rocky headlands at Cabarita Beach. Seeds belonging to 20 families of each Dune and Headland ecotype were sown into 16 trays (each family was represented by one seed per tray) for a total of 640 seeds. Seeds were sown on top of the soil in a fully randomized design within each tray. Soil was sprayed with water three times a day to keep it moist, and tray position on the shelf was switched daily. Room temperature was at 25°C through a 12 h : 12 h, light : dark cycle. Seed germination and death occurred during the first 19 d of the experiment. A nominal logistic mixed model (GLM, Generalized Linear Model for binary data) was used to analyze the proportion of germination for each cross type (including parental and hybrid crosses) within soil, using the *lmer* package in R (R Development Core Team, 2012). The model included cross type as a fixed effect, and tray and family as random effects.

In order to further study the germination of Dune and Headland ecotypes from Cabarita Beach at other localities where Dune and Headland populations also grow, we conducted a second soil experiment. We used soil from three different sites (two in addition to Cabarita Beach) collected from the sand dunes and rocky headlands at Lennox Head and Stradbroke Island. We filled two trays per locality for a total of six trays. Using the same set-up conditions as the experiment above, we sowed a total of 420 seeds (35 seeds for five families of each ecotype per tray). Germination and mortality were recorded daily during the first 33 d after which no more germination or death occurred. Statistical analyses were performed using a GLM to test for the full interaction among soil, locality and cross type. We also performed an analysis within soil to test for germination differences between ecotypes. Cross type and soil were treated as fixed effects, and tray and family as random effects.

Reciprocal transplant experiment in the field A total of 1760 seeds belonging to 70 families of Dune, 70 of Headland and 80 families of reciprocal F_1s (F1-D and F1-H correspond to F_1 individuals with either a Dune or Headland mother, respectively) were sown directly into the field in four plots in both the Dune and Headland environments (one seed per family per plot in a fully randomized design) in November 2010. To track seeds in the field, they were individually pasted at their mid-point onto toothpicks using Selleys Parfix superglue (ensuring neither extreme of the seeds was covered that would have obstructed germination). Control experiments in the laboratory and other reciprocal transplant experiments (not shown here) have shown that gluing seeds to toothpicks does not affect germination rates. Seeds were sown 5 mm beneath the soil surface. Plots were covered with a 50% UV protection mesh (HDPE UV stabilized

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forest green exterior fabric) to prevent loss of seeds during the rainy season. Seeds were lightly sprayed with water once a day for the first 2 wk of the experiment to keep the surface of the soil moist. Experiments were held during the wet season and an average of 313.69 mm of rainfall fell during the first 45 d of the experiment.

Germination and mortality were recorded twice a day for the first month. Seedlings were considered killed by herbivores if only their stalks were found after two consecutive survivorship measurements. We termed predators any invertebrates that were found on seedlings which ate both cotyledons, leading to plant death. The few plants that disappeared between two consecutive measurements were also considered killed by predators. Analyses with and without these plants did not change the interpretation of results. Plants that wilted and died slowly were considered to have died by 'other causes', likely involving drought and predation. Because these individuals died slowly (i.e. progressive desiccation of the seedling), we were able to monitor death progression over several survivorship measurements. Survivorship data were taken over several hours during the morning and afternoon of each day of the first month, when most individuals died. For the following 4 months we visited the site weekly, and for the rest of the experiment every second week. A GLM was used to analyze the proportion of either germination or death due to predation for crosses within environment; the model included cross type as a fixed effect, and family and block as random effects. To further evaluate the effects of cross type on total survivorship, we performed a GLM using the Poisson distribution (to account for the large proportion of individuals that died in the two environments) on the number of days alive. We censored the plants up to when the first flower was produced in each habitat (sand dunes day 275 and rocky headland day 149). Standard survivorship analyses (Fox, 2001) did not affect the conclusions derived from our results. Finally, we conducted a GLM to test the effect of parental and hybrid genotypes on the average number of flowers in each environment. We fitted general linear models using restricted maximum likelihood in the *lmer* package in R.

Strength of RI

We calculated the strength of several intrinsic (*I*) and extrinsic (*E*) RI barriers between the two coastal ecotypes, following the approach of Lowry *et al.* (2008a). We calculated the following reproductive barriers: (1) *Flowering asynchrony in the field*, as $RI_{phen1} = 1 - (\text{observed/expected interpopulation matings})/(\text{observed/expected intrapopulation matings})$. (2) *Immigrant inviability* (*E*), or whether migrant seeds had difficulties establishing in the alternative ecotype environment, as $RI_{imm} = 1 - (w_i/w_n)$, where w_i is the mean number of surviving migrant individuals, and w_n the surviving local type. (3) *Hybrid viability* (*I*, *E*), or whether hybrid seedlings germinated and survived equally well as their parents in field or controlled conditions, as $H_{hf} = 1 - (v_{\text{mean}}F_1/v_{\text{local}})$, where v_{mean} is the average survivorship for F_1 hybrids, and v_{local} is the average survivorship of the local ecotype. In the field, we also estimated hybrid viability by only taking into account the mortality of individuals that germinated, thus disentangling the effects of lack of hybrid seed germination from hybrid mortality. (4) *Hybrid seed set* (under controlled conditions) (*I*) or whether the proportion of fertilized seeds in a flower head from an inter-ecotype cross differed from an intra-ecotype cross $RI_{\text{seed set}} = 1 - (\text{Pf}_{\text{inter}}/\text{Pf}_{\text{intra}})$, where Pf stands for proportion fertilized. We estimated the total cumulative RI for each ecotype taking into account the absolute contribution of each reproductive barrier in the study according to the methodology in Lowry *et al.* (2008a). Finally, we calculated the magnitude of local adaptation (local adaptation index) for each of the crosses as described in Hereford (2009). Three estimates of local adaptation were obtained depending on the fitness measure taken into account: average number of days in which each cross type was alive until the end of the experiment, average number of flowers per individual, and the product of these two variables.

Results

Ecotypic differences

In Cabarita Beach, Dune (D) individuals were always erect or decumbent, whereas all Headland (H) individuals were short and prostrate. In the glasshouse, H individuals – with one exception – were prostrate, and all D individuals were erect or decumbent ($F_{1,37} = 33.5013$, $P < 0.0001$). The H population also had individuals with more branches than the D population ($F_{1,37} = 28.6556$, $P < 0.0001$). In the field, the D population flowered little or did not flower from November to March, while the H population flowered throughout the entire year (Fig. 1). In the glasshouse, H plants flowered after 8 wk, while D individuals flowered after 10 wk. In the glasshouse, both populations flowered for *c.* 4 months, after which plants stopped producing new leaves or flowers. Consistent with previous reports for the system (Ali, 1968; Radford *et al.*, 2004), growth habit differences between Cabarita Beach populations were retained in the

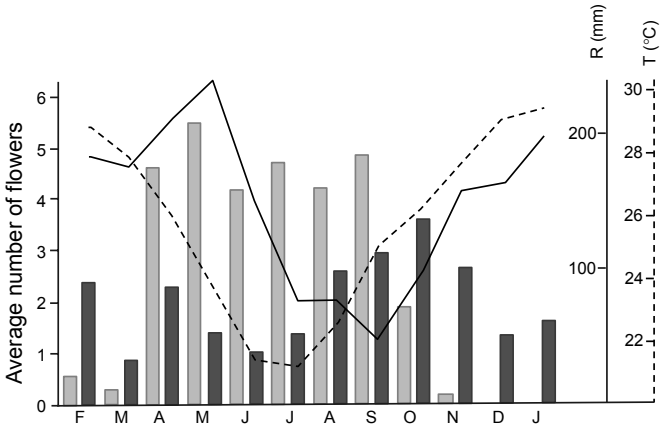


Fig. 1 Average number of flowers per individual from the natural populations of *Senecio lautus* in the sand dunes (light gray bars) and rocky headlands (dark gray bars) through 12 consecutive months (February 2011 to January 2012). Monthly average rainfall (R; mm) (solid line) and temperature (T) in °C (dashed line) are averages over the past 10 yr.

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glasshouse, but flowering time was affected by the environment in which they grew.

Intrinsic RI

Seeds produced in the glasshouse from crosses within or between populations were highly viable (germination success in moist filter paper >98% for both parents and hybrids). Hybrid and parental seed set was similar (mean seed set: H = 0.45 ± 0.08; D = 0.48 ± 0.08, F_1 = 0.49 ± 0.07, $F_{2,62}$ = 0.0825, P = 0.9358) indicating that intrinsic RI is weak between the D and H populations at Cabarita Beach. Intrinsic RI in other population pairs was generally weak (Table 1). The population pair at Hat Head showed the greatest level of RI, but only in one direction of the cross. We found negative values of RI in some crosses, particularly in the cross between a tableland and a headland population. Overall, F_1 hybrid fitness is similar to parental fitness in the *S. lautus* populations studied here.

Extrinsic RI

Soil experiments in the glasshouse Although overall germination success in the glasshouse was low, parental seeds germinated equally well in both sandy and rocky soils from Cabarita Beach (D soil, z = −1.732, P = 0.0834; H soil, z = −1.059, P = 0.290; Fig. S3, Table S2). The H and D seeds sown in soil collected from other dune and headland localities showed variable patterns of germination success (interaction model, χ^2 = 36.807, P < 0.0001; Fig. 2, Table S2). Germination differences were most pronounced in Lennox Head soil (D soil, z = −2.794, P = 0.005; H soil, z = 2.942, P = 0.003) but less apparent and asymmetric in Stradbroke Island soil (D soil, z = −1.938, P = 0.053; H soil, z = −0.549 P = 0.583). Patterns of germination success did not change in Cabarita Beach soil (D soil, z = −0.804, P = 0.421; H soil, z = −0.305, P = 0.760; note that this is a replicate of the experiment already described).

Table 1 Intrinsic reproductive isolation (RI) in *Senecio lautus* measured in the glasshouse

-patry	Locality	Cross[a]	N[b]	RI[c]
Para	Hat Head	H × D	25	−0.053
		D × H	25	0.241
	Lennox Head	H × D	54	−0.145
		D × H	60	0.098
	Cabarita Beach	H × D	25	−0.089
		D × H	25	0.015
Allo	Cabarita Beach/Byron bay	D × H	27	0.056
		H × D	27	−0.221
	Lamington NP/Port Macquarie	T × H	20	−0.230
		H × T	20	−0.166

[a]D, H and T refer to Dune, Headland and Tableland population.
[b]N is the number of families that were crossed.
[c]RI is the strength of postmating prezygotic RI measured as the relative fecundity of hybrids (F_1 seed set) over that of parental types (parental seed set) in three parapatric and two allopatric populations.

Reciprocal transplant experiments in the field – germination and mortality The overall germination success in the field was low (Fig. 3a, Table S2), particularly in the H environment (z = −9.279, P < 0.001), but did not differ between parental crosses in either of the two habitats (sand dunes, z = −0.913, P = 0.361, rocky headland, z = −1.673, P = 0.0943). By contrast, F_1 hybrids germinated with significantly greater success than either parent in the field, particularly in the sand dunes (z = 3.137, P = 0.0017), and when they carried a D cytoplasm (z = 3.973, P < 0.001). Germination occurred in two independent bouts, but this did not affect the overall pattern of germination success between crosses and in either environment (Table 2). Censored adult survivorship differed between crosses in both environments (Fig. 4, Table S3). In the sand dunes, D families survived better than other crosses (H, z = −5.847, P < 0.001; F1-H, z = −3.158, P = 0.0015 and F1-D, z = −3.035, P = 0.0024). In the rocky headland, where selection was strongest, H families showed the highest survivorship, although differences were not significant amongst cross types (D, z = −1.396, P = 0.1630; F1-D, z = 0.459, P = 0.6460 and F1-H, P = 0.7820). Overall, the combined effects of germination and mortality indicate that the demography of transplanted populations is qualitatively different between environments (Fig. 3b). Thus, when the proportion of

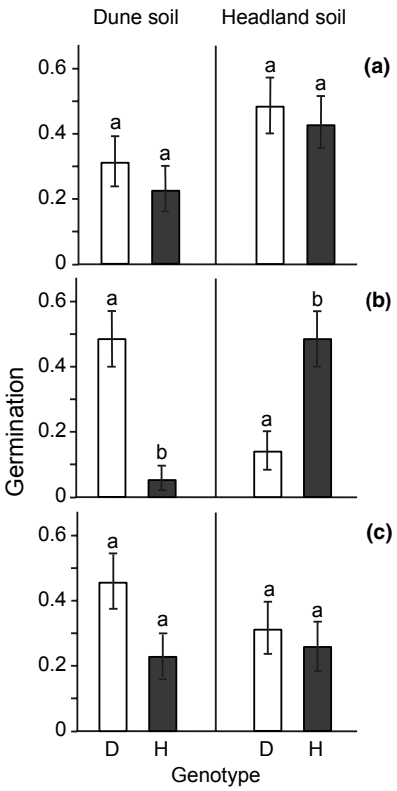


Fig. 2 Proportion of *Senecio lautus* Dune (D) and Headland (H) seeds that germinated in D soil (left panels) and H soil (right panels) collected from three different localities: (a) Cabarita Beach, (b) Lennox Head and (c) Stradbroke Island. Bars show standard errors with letters denoting significant differences. These experiments were conducted under controlled environmental conditions in the glasshouse.

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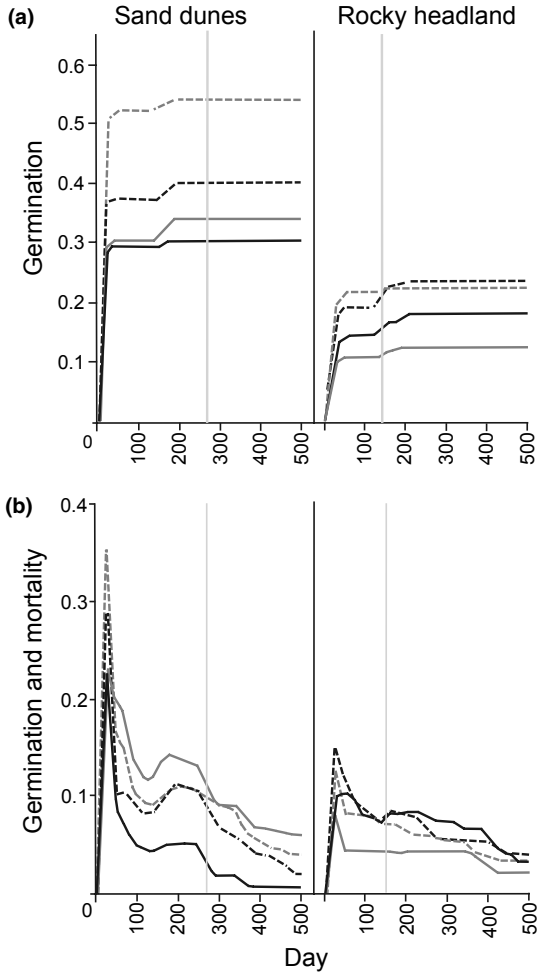


Fig. 3 Germination and mortality for parental and hybrid types of *Senecio lautus* populations in field transplant experiments. (a) Proportion of seeds that germinated in experimental blocks in the sand dunes and rocky headland of Cabarita Beach. (b) Combined effects of germination and mortality on the total number of survivors relative to the total number of seeds planted. Vertical lines indicate the day at which the first flower appeared in each environment (day 275 in the sand dunes and day 149 in the rocky headland). Black and gray lines correspond to the Headland and Dune crosses, respectively, and the dashed black and gray lines to F1-H (Headland mother) and F1-D (Dune mother) crosses, respectively.

Table 2 Proportion of germination for each experimental cross in reciprocal transplant experiments in Cabarita Beach using Dune and Headland populations of *Senecio lautus*

Habitat	Cross type[a]	*N*	G1[b]	G2[c]
Dune	D	275	30.59	34.18
	F1-D	159	52.12	54.08
	F1-H	159	37.15	40.25
	H	269	29.38	30.48
Headland	D	271	10.75	12.54
	F1-D	151	21.89	22.51
	F1-H	153	18.98	23.52
	H	265	14.78	18.11

[a]Cross types are Dune (D) and Headland (H) parental types and reciprocal F_1 hybrids where letters denote the identity of the mother (F1-H and F1-D).
[b]G1 is the proportion of seeds that germinated during the first days of the experiment.
[c]G2 refers to a second bout of germination (day 135 in the headland environment, and day 145 in the dune environment).

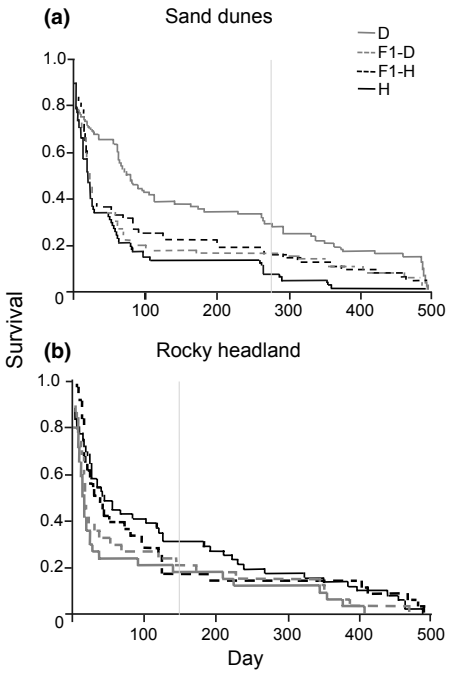


Fig. 4 Proportion of *Senecio lautus* parental and hybrid types alive (survival) in the (a) sand dunes and (b) rocky headland of Cabarita Beach. Vertical lines indicate the day at which the first flower appeared at each environment (day 275 in the sand dunes and day 149 in the rocky headland). D, individuals from the Dune cross; F1-H, those from the hybrid cross where the mother was a Headland individual; F1-D, those from the hybrid cross where the mother was a Dune individual; and H, individuals from the Headland cross.

individuals that had germinated and survived was plotted throughout the course of the experiment (i.e. until most of the population died; Fig. 3b) it was evident that (1) germinated seed of all types was subject to mortality, although absolute mortality was greater for F_1s, but intermediate between the two parental types; (2) the local type (D or H in its local environment) performed better across the duration of the experiment (see fecundity results below); (3) by the end of the experiment (*c.* 500 d) a proportion of each type of seed had germinated and produced individuals that had completed their life cycle (Fig. 3b); and (4) the total number of individuals alive at the first day of flowering (see gray vertical line in each panel of Fig. 3b) in each environment differed amongst cross type and in the direction of local adaptation ($\chi^2 = 16.3$, df = 3, $P = 0.001$; Table 3).

Reciprocal transplant experiments in the field – mortality due to herbivory We found herbivores on the seedlings planted at Cabarita Beach (e.g. Fig. S4). Because herbivores killed plants, we refer to this process as predation. Death due to predation occurred only when seedlings had green cotyledons (once seedlings produced new leaves, we detected mortality events due to

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Table 3 Individuals with flowers and their average number of flowers for each experimental cross in reciprocal transplant experiments in Cabarita Beach using Dune and Headland populations of *Senecio lautus*

Habitat	Cross type[a]	Seeds sowed	Alive at first day of flowering	Mean number of flowers per individual	SE
Dune	D	275	28	0.0641	0.0447
	F1-D	159	15	0.0074	0.0073
	F1-H	159	13	0	0
	H	269	6	0	0
Headland	D	271	8	2.6475	2.2360
	F1-D	151	8	1.1260	0.5569
	F1-H	153	12	2.0125	1.1435
	H	265	17	1.0425	0.5325

[a]Cross types are Dune (D) and Headland (H) parental types and reciprocal F_1 hybrids where letters denote the identity of the mother (F1-H and F1-D).

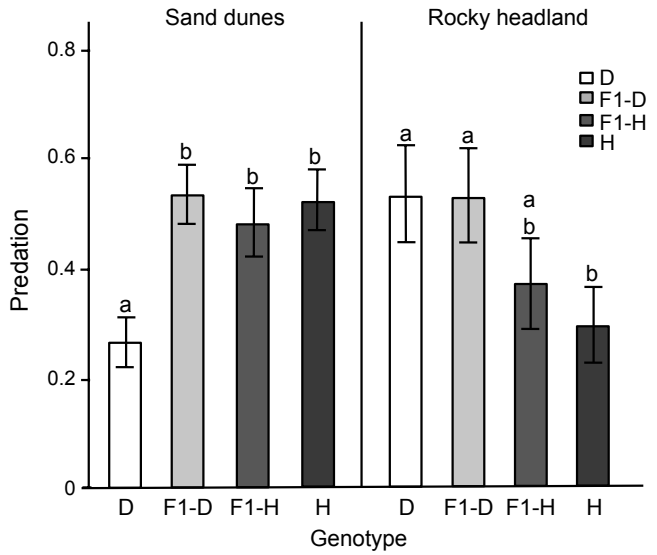


Fig. 5 Proportion of *Senecio lautus* seedlings that die because of herbivory (predation) in the sand dunes and rocky headland of Cabarita Beach. Bars show standard errors for binomial probabilities, with letters denoting significant differences. Bars ranging in color from white to dark gray represent the Dune, F1-D, F1-H, and Headland cross types, correspondingly.

other causes, possibly drought). In both habitats, herbivores killed immigrants more often than local parents (sand dunes, $z = 3.753$, $P = 0.0001$; rocky headland, $z = 2.159$, $P = 0.0309$; Fig. 5, Table S3). In both environments, hybrids were attacked significantly more than the local type, with the H cytoplasm conferring a slight advantage in the rocky headland (sand dunes, F1-D, $z = 3.763$, $P = 0.0002$; F1-H, $z = 2.904$, $P = 0.0037$; rocky headland, F1-D, $z = 2.124$, $P = 0.0337$, F1-H, $z = 0.766$, $P = 0.4434$; Fig. 5, Table S3 for proportions of predated individuals). Crosses did not show differences in mortality due to 'other possible causes' of death in any of the two environments (sand dunes, H, $z = 0.235$, $P = 0.8139$; F1-D, $z = -0.404$, $P = 0.6863$; F1-H, $z = -0.017$, $P = 0.9863$; rocky headland, D, $z = 0.103$, $P = 0.9181$; F1-D, $z = -0.038$ $P = 0.9699$, F1-H, $z = -0.186$, $P = 0.8526$).

Reciprocal transplant experiments in the field – number of flowers The first flower head buds on D plants appeared after 275 d in the sand dunes and after 191 d in the rocky headland. By contrast, H individuals never flowered in the sand dunes, but flowered after 191 d in the rocky headland. The average number of flowers heads per individual (including those that did not flower) was randomly distributed with respect to cross type (sand dunes, $F_{3,79} = 0.779$, $P = 0.5092$, rocky headland, $F_{3,42} = 0.347$, $P = 0.7912$; Table 3).

Please refer to Tables S2 and S3 for means and SE for all fitness components measured in each experiment, and see Table S4 for a combined summary of all statistical tests performed.

Strength of RI

For the barriers we measured, cumulative RI for Cabarita Beach populations was 0.88 in the sand dunes, with average prezygotic and postzygotic strength of 0.59 and 0.11, respectively. In the headland, cumulative RI was 0.76, with average prezygotic and postzygotic strength of 0.55 and 0.04, respectively. Overall, intrinsic reproductive barriers were absent (Table 1), whereas extrinsic reproductive barriers were strong (Table 4). In particular

Table 4 Strength of reproductive isolation (RI) barriers between Dune and Headland populations of *Senecio lautus* found at Cabarita beach

Reproductive isolating barrier[a]	Habitat	Cross[b]	Strength
Flowering asynchrony	Dune	–	0.3939
	Headland	–	0.48
Immigrant inviability	Dune	–	0.78
	Headland	–	0.54
Extrinsic postzygotic[c]	Dune	F1-D	0.41
		F1-H	0.32
		F_1	0.37
	Headland	F1-D	0.34
		F1-H	0.06
		F_1	0.19
Hybrid F_1 seed set	Glasshouse	F1-D	0.015
		F1-H	−0.089
		F_1	−0.038
Intrinsic hybrid viability	Glasshouse	F1-D	0
		F1-H	0
		F_1	0
Average prezygotic	Dune		0.59
Average postzygotic			0.11
Cumulative			0.877
Average prezygotic	Headland		0.55
Average postzygotic			0.038
Cumulative			0.76

[a]Estimates were calculated following Lowry *et al.* (2008a) approach.
[b]Cross types are Dune (D) and Headland (H) parental types and reciprocal F_1 hybrids where letters denote the identity of the mother (F1-H and F1-D).
[c]Extrinsic postzygotic isolation was calculated from either the total number of seeds or from individuals that germinated in the reciprocal transplant experiment, respectively.

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Table 5 Magnitude of local adaptation (local adaptation index) at each habitat for each experimental cross in reciprocal transplant experiments in Cabarita Beach using Dune and Headland populations of *Senecio lautus*

Habitat	Local vs nonlocal [a] cross type	Magnitude of local adaptation		
		Viability	Fecundity	Composite
Dune	D vs H	0.96	2.56	3.05
	D vs F1-H	0.61	2.56	3.05
	D vs F1-D	0.70	1.12	2.10
Headland	H vs D	0.55	−1.08	−0.35
	H vs F1-D	0.40	0.07	0.44
	H vs F1-H	0.23	−0.40	−0.19

[a]Cross types are Dune (D) and Headland (H) parental types and reciprocal F_1 hybrids where letters denote the identity of the mother (F1-H and F1-D).

immigrant inviability and extrinsic postzygotic isolation seemed to play a major, albeit asymmetric, role in preventing gene flow between the two populations. Positive indexes of local adaptation, as measured by Hereford (2009), showed that – when viability was the measure of fitness – each ecotype performed best in its own environment and F_1 performance was intermediate but slightly better depending on whether they carried the local cytoplasm (Tables 5, and S4 for further details). This was not the case when fitness measurements included number of flowers, although these measurements relied on small population sizes during the flowering season.

Discussion

We have shown that extrinsic reproductive isolation (RI) creates strong barriers to gene flow between two neighboring coastal populations of the Australian groundsel *Senecio lautus*. These populations show morphological differences with a strong genetic basis that persist despite constraints of gene flow on divergence (Roda *et al.*, 2013a). We discovered that predators were the main cause of divergent natural selection and they contributed to selection against both migrants and hybrids between parental populations. Finally, and in contrast to several previous studies connecting local adaptation and the origins of RI in plants (reviewed in Lowry *et al.*, 2008a), we found that although F_1 hybrids germinated and established initially much more than parents, they eventually suffered more individual losses to predators and other causes (possibly drought) than their parents. The fact that F_1 hybrids showed hybrid vigor during the initial stages of development, suggests that heterosis and development may interact during the evolution of extrinsic RI. Below we discuss these main findings and their implications for understanding the progress towards speciation and the origins of RI in plants.

Lack of intrinsic RI in the F_1 generation

We did not detect intrinsic RI between the two populations at Cabarita Beach, consistent with our measures of F_1 seed set between allopatric pairs (Table 1) and with previous reports in the system where most populations were easily crossed and pollen fertility was generally high (Ornduff, 1964). Nevertheless, it is possible that hybrid fitness is reduced in later generations (i.e. hybrid breakdown), a common phenomenon detected in many plants (Fishman & Willis, 2001; Moyle & Graham, 2005). Although we did not measure F_2 hybrid fitness between Cabarita Beach populations, we have unpublished data from other *S. lautus* populations where we have found hybrid breakdown, possibly suggesting that Dobzhansky–Muller incompatibilities (e.g. dominant × recessive, and recessive × recessive) have accumulated in the system. It is also possible that other intrinsic barriers such as conspecific pollen precedence might play an important role in this system, particularly when insect communities are shared across coastal habitats (White, 2008; Fig. S5).

Absence of some forms of intrinsic RI between populations adapting to contrasting environments is not uncommon, and may be normal during the early stages of speciation (Schluter, 1998; Hendry *et al.*, 2007), although there are clear examples where the two evolve together (Macnair & Christie, 1983). For instance, species adapted to serpentine soil lack intrinsic RI, yet they persist in parapatry, and sometimes completely cease exchanging genes (Brady *et al.*, 2005; Harrison & Rajakaruna, 2011). Similarly, inland and coastal populations of *Mimulus guttatus* show weak intrinsic RI, yet they are largely reproductively isolated from one another (Lowry *et al.*, 2008b). Finally, invertebrate (Via *et al.*, 2000; Nosil, 2007; McBride & Singer, 2010) and vertebrate (Hatfield & Schluter, 1999; Fuller *et al.*, 2007) species have populations that display strong morphological differentiation but weak intrinsic RI. Whether extrinsic barriers to gene flow can be considered triggers of speciation requires further work (Nosil *et al.*, 2009) but simulations have shown that strong selection against migrants is an effective and rapid way to reduce gene flow between populations (Hendry, 2004; Thibert-Plante & Hendry, 2009).

Heterotic and cytoplasmic effects on extrinsic RI

In our field experiments we found strong differences between parental and hybrid types in germination and survival, contrary to glasshouse results (Fig. 3). Germination success was significantly higher in hybrids compared with parental types, consistent with other studies where hybrids show heterotic effects (reviewed in Lowry *et al.*, 2008a; Sambatti *et al.*, 2012). However, we detected this heterotic effect only during the early life cycle stages, suggesting an influence of parental cytoplasmic effects (i.e. the mother of the F_1 hybrid had effects on its germination success in the field; Figs 3 and 5, and see Discussion later). Although reasons for the initial extrinsic heterotic effect remain unclear, a release of antagonistic effects in hybrids (Burke & Arnold, 2001) might partially explain our observations: where genes controlling growth and reproduction no longer function together at later stages of development.

Our field experiments suggest that there could be extrinsic cytoplasmic effects on hybrid fitness. This result echoes those found on *Ipomopsis aggregata* and *I. tenuituba* (Campbell & Waser, 2001) and in *Chamaecrista fasciculata* (Galloway &

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Fenster, 1999) where a similar distinction in hybrid fitness between benign and field conditions were found. The reason for these effects is unknown, but it could be the result of stress dependent cyto-nuclear incompatibilities (Coyne & Orr, 2004). In our field experiment, F_1 hybrids showed mortality patterns consistent with accumulation of stress through development: F_1 individuals displayed hybrid vigor during early development but failed to survive to late developmental stages. However, most deaths were due to predation, so for stress to be a viable hypothesis, we must predict that predators preferred stressed to non-stressed hybrids. Although this possibility deserves further experimentation, two observations suggest that stress alone is not the main cause: first, predators ate healthy plants and not wilted plants; and second, previous studies have found that some phytophagous insects frequently prefer healthy plants over water stressed ones, as water stress can interfere with their ability to avail nitrogen (Huberty & Denno, 2004). Field experiments with cross types containing an increasing proportion of a local genome (e.g. reciprocal backcrosses, F_1, and F_2 hybrids) could help disentangle environmentally-dependent and intrinsic causes of mortality in this system (Rundle & Whitlock, 2001). Maternal provisioning of nutrients to seeds could also affect hybrid performance, but adding seed mass as a covariate in our analyses did not affect germination or survivorship (data not shown).

Predation creates extrinsic RI

Systems where immigrant inviability is strong are suggestive of strong extrinsic postzygotic RI (Lowry *et al.*, 2008a). Although this has been demonstrated in a few cases in animals (Via *et al.*, 2000), such a link is less clear in plant systems (Lowry *et al.*, 2008a). For instance, in studies of the sister species *Ipomopsis agregata* and *I. tenuitubai*, selection against migrants was noticeable, but F_1s performed on average similarly to parental types, although there were reductions in survival depending on the direction of the cross (Campbell & Waser, 2001). Similarly, in *Artemisia tridentata* subspecies, hybrids displayed a fitness advantage in most habitats in which they grew (Miglia *et al.*, 2005), thus possibly facilitating the opportunity of gene flow between subspecies. However, manipulative experiments in some plants have demonstrated that selection against migrants and hybrids could evolve quickly (Jain & Bradshaw, 1966; Davies & Snaydon, 1976; Hendry *et al.*, 2007).

In our experiments, herbivory created both extrinsic prezygotic and postzygotic RI barriers between parapatric populations of *S. lautus* (Table 4). Previous studies report that herbivores partially consumed fractions of leaves or flowers of adult individuals (Combes, 1996; Fritz *et al.*, 1999); however, our results are more related to animal examples where individuals were killed by attacks from other organisms (Nosil, 2004; Langerhans *et al.*, 2007) due to predation of newly emerged seedlings resulting in deaths. Although we cannot currently explain the causative agents for differential predation on parental vs migrant and hybrid seedlings, studies on other species in the *Senecio* genus have revealed that toxic alkaloids may serve as plant defenses against insects, and that production of such alkaloids is largely dependent on the environment where the species occurs (Kirk *et al.*, 2010). It would not be surprising if divergence in the type and amount of alkaloids in the cotyledons of *S. lautus* seedlings were responsible for the evolution of extrinsic RI between ecotypes. However, further studies on the genetics of immigrant and extrinsic postzygotic RI are required to further understand how they can evolve concurrently. Overall, our results link proximate causes (predation) with ultimate causes of divergence (natural selection; Laland *et al.*, 2011), and suggest that natural selection reduces the exchange of genes between the Dune (D) and the Headland (H) ecotypes through the formation of maladapted hybrids and selection against migrants.

Although predation is an important agent of selection, other environmental variables could contribute to divergence between D and H populations. For instance, a previous study in the *Senecio* system found that soil composition differed drastically between D and H environments, and with large localized variation (Roda *et al.*, 2013b). Furthermore, a large proportion of allelic variation in D and H parapatric pairs, including Cabarita Beach, correlated with variation in abiotic elements found in their soils, including salt, metal and nutrient content (Roda *et al.*, 2013b). These results suggested that soil content contributes to adaptation to sand dunes and rocky headlands and may partially explain why we saw variable germination rates between the dune and headland environments (Fig. 2).

Stages of speciation

Studies of speciation have considered it useful to treat ecotypes as a stage of the process of species formation (Clausen *et al.*, 1947; Lowry, 2012). This view can help us understand the role of ecology on speciation and the relative contributions of reproductive isolating barriers to each stage of the process. The mechanisms facilitating transitions between ecotype and species remain mysterious, although it is possible that once extrinsic barriers to gene flow establish, neutral differentiation accumulates, thus leading to the evolution of genetic incompatibilities responsible for various forms of intrinsic RI in a system (Nosil *et al.*, 2008). Additionally, general extrinsic barriers to gene flow may facilitate novel sweeps in each population if the loci under selection are linked to those under initial divergent selection (Hendry *et al.*, 2007). This form of selection may be powerful and fast in creating further barriers to gene flow including intrinsic ones. Likewise, environmental effects on the time of reproduction (e.g. flowering time differences were marked in the natural populations in the field but limited in the glasshouse) may reduce gene flow even in the absence of genetic divergence between populations and thus promote the subsequent evolution of genetically based reproductive barriers (e.g. Thibert-Plante & Hendry, 2011). Finally, gamete competition within populations could lead to conspecific pollen precedence between populations, perhaps rapidly driving the evolution of strong barriers to gene flow during the early stages of speciation (Howard, 1999).

Overall, Cabarita Beach populations have evolved extrinsic barriers to gene flow, and the strength of RI is relatively high (see the Results section and Table 4). According to the classification

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of stages in the progress toward speciation in Hendry *et al.* (2009) – that ranges from totally panmictic populations to completely and irreversibly isolated species – D and H populations at Cabarita Beach are ecotypes that seem to be in an intermediate stage of divergence: 'Strongly discontinuous variation between populations with strong but reversible RI' (Hendry *et al.*, 2009). Whether the D and H populations will become discrete species is not currently possible to know, but our recent multi-locus estimates of divergence between multiple coastal population pairs suggest that divergent natural selection can take populations to varying degrees of divergence in some cases with no detectable levels of gene flow from molecular markers (M. C. Melo *et al.*, unpublished results). Multiple D and H parapatric pairs in *S. lautus* could help us to identify the factors affecting (favoring/constraining) the progress toward ecological speciation and thus inform us about the different points at which the distinct pairs could be in the ecological speciation continuum (Hendry *et al.*, 2009; Nosil *et al.*, 2009). We expect that studies on the genetic basis of adaptation in this system will shed light as to whether regions responsible for extrinsic RI could persist.

Conclusions

Field and glasshouse experiments described here suggest that both immigrant inviability and extrinsic postzygotic isolation create strong barriers to gene flow between D and H populations of *S. lautus*. In agreement with other studies of diverging populations adapted to contrasting habitats (Clausen *et al.*, 1947; McNeilly & Antonovics, 1968; reviewed in Lowry *et al.*, 2008b), immigrant inviability was stronger than hybrid inviability (Table 4; Ramsey *et al.*, 2003; Nosil *et al.*, 2005), intrinsic barriers were barely noticeable in F_1 hybrids, and natural selection (local adaptation index) was most effective before the reproductive season, in particular, at the seedling stage. In general, our results resemble those found in animal systems such as *Timema* walking sticks (Nosil, 2004), sticklebacks (Hatfield & Schluter, 1999) and pea aphids (Via *et al.*, 2000), but contrast with results found in other plant systems such as *Mimulus*, where F_1 extrinsic postzygotic reproductive isolation between ecotypes is rather weak (Lowry *et al.*, 2008b), but strong when locus specific effects are considered in later hybrid generations (Lowry & Willis, 2010). We suggest that ecological reproductive isolation between plant and animals may follow similar paths.

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References

Abbott RJ. 1976. Variation within common groundsel, *Senecio vulgaris* L. I. Genetic response to spatial variations of the environment. *New Phytologist* **76**: 153–164.

Ali S. 1968. *Senecio lautus* complex in Australia. IV. The biology of the complex. *Phyton (Horn, Austria)* **13**: 53–62.

Ali SI. 1964. *Senecio lautus* complex in Australia. I. Taxonomic considerations and discussion of some of the related taxa from New Zealand. *Australian Journal of Botany* **12**: 282–291.

Antonovics J. 2006. Evolution in closely adjacent plant populations X: long-term persistence of preRI at a mine boundary. *Heredity* **97**: 33–37.

Antonovics J, Bradshaw AD. 1970. Evolution in closely adjacent plant populations. VIII. Clinal patterns at a mine boundary. *Heredity* **25**: 349–362.

Arnold ML. 1997. *Natural hybridization and evolution*. New York, NY, USA: Oxford University Press.

Barton N. 2001. The role of hybridization in evolution. *Molecular Ecology* **10**: 551–568.

Barton NH, Hewitt G. 1985. Analysis of hybrid zones. *Annual Review of Ecology and Systematics* **16**: 113–148.

Berglund ABN, Dahlgren S, Westerbergh A. 2004. Evidence for parallel evolution and site-specific selection of serpentine tolerance in *Cerastium alpinum* during the colonization of Scandinavia. *New Phytologist* **161**: 199–209.

Berner D, Kaeuffer R, Grandchamp AC, Raeymaekers J, Räsänen K, Hendry A. 2011. Quantitative genetic inheritance of morphological divergence in a lake–stream stickleback ecotype pair: implications for reproductive isolation. *Journal of Evolutionary Biology* **24**: 1975–1983.

Bordenstein S, Drapeau M. 2001. Genotype-by-environment interaction and the Dobzhansky-Muller model of postzygotic isolation. *Journal of Evolutionary Biology* **14**: 490–501.

Bradshaw H, Schemske DW. 2003. Allele substitution at a flower colour locus produces a pollinator shift in monkeyflowers. *Nature* **426**: 176–178.

Brady KU, Kruckeberg AR, Bradshaw H Jr. 2005. Evolutionary ecology of plant adaptation to serpentine soils. *Annual Review of Ecology, Evolution, and Systematics* **36**: 243–266.

Bray EA. 2002. Classification of genes differentially expressed during water-deficit stress in *Arabidopsis thaliana*: an analysis using microarray and differential expression data. *Annals of Botany* **89**: 803–811.

Burke JM, Arnold ML. 2001. Genetics and the fitness of hybrids. *Annual Review of Genetics* **35**: 31–52.

Campbell DR, Waser NM. 2001. Genotype-by-environment interactions and the fitness of plant hybrids in the wild. *Evolution* **55**: 669–676.

Clausen J, Keck DD, Hiesey WM. 1947. Heredity of geographically and ecologically isolated races. *American Naturalist* **81**: 114–133.

Combes C. 1996. Parasites, biodiversity and ecosystem stability. *Biodiversity and Conservation* **5**: 953–962.

Coyne JA, Orr HA. 2004. *Speciation*. Sunderland, MA, USA: Sinauer Associates.

Czesak M, Knee M, Gale R, Bodach S, Fritz R. 2004. Genetic architecture of resistance to aphids and mites in a willow hybrid system. *Heredity* **93**: 619–626.

Davies M, Snaydon R. 1976. Rapid population differentiation in a mosaic environment. *Heredity* **36**: 59–66.

DellíOlivo A, Hoballah ME, Gübitz T, Kuhlemeier C 2011. Isolation barriers between *Petunia axillaris* and *Petunia integrifolia* (Solanaceae). *Evolution* **65**: 1979–1991.

Elias M, Faria R, Gompert Z, Hendry AP. 2012. Factors influencing progress toward ecological speciation. *International Journal of Ecology* 2012: Article ID 235010.

Emms S, Arnold M. 1997. The effect of habitat on parental and hybrid fitness: transplant experiments with Louisiana irises. *Evolution* **51**: 1112–1119.

Felsenstein J. 1981. Skepticism towards Santa Rosalia, or why are there so few kinds of animals? *Evolution* **35**: 124–138.

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Fishman L, Willis JH. 2001. Evidence for Dobzhansky-Muller incompatibilities contributing to the sterility of hybrids between *Mimulus gutatus* and *M. nasutus*. *Evolution* **55**: 1932–1942.

Fox GA. 2001. Failure-time analysis. In: Scheiner SM,Gurevitch J, eds. *Design and analysis of ecological experiments*. New York, NY, USA: Oxford University Press, 253–289.

Fritz R, Moulia C, Newcombe G. 1999. Resistance of hybrid plants and animals to herbivores, pathogens, and parasites. *Annual Review of Ecology and Systematics* **30**: 565–591.

Fritz R, Nichols-Orians C, Brunsfeld S. 1994. Interspecific hybridization of plants and resistance to herbivores: hypotheses, genetics, and variable responses in a diverse herbivore community. *Oecologia* **97**: 106–117.

Fuller R, McGhee K, Schrader M. 2007. Speciation in killifish and the role of salt tolerance. *Journal of Evolutionary Biology* **20**: 1962–1975.

Galloway LF, Fenster CB. 1999. The effect of nuclear and cytoplasmic genes on fitness and local adaptation in an annual legume, *Chamaecrista fasciculata*. *Evolution* **53**: 1734–1743.

Gavrilets S, Li H, Vose MD. 2000. Patterns of parapatric speciation. *Evolution* **54**: 1126–1134.

Harrison S, Rajakaruna N. 2011. *Serpentine: the evolution and ecology of a model system*. Berkeley, CA, USA: University of California Press.

Hatfield T, Schluter D. 1999. Ecological speciation in sticklebacks: environment-dependent hybrid fitness. *Evolution* **53**: 866–873.

Hendry A. 2004. Selection against migrants contributes to the rapid evolution of ecologically dependent reproductive isolation. *Evolutionary Ecology Research* **6**: 1219–1236.

Hendry A, Bolnick D, Berner D, Peichel C. 2009. Along the speciation continuum in sticklebacks. *Journal of Fish Biology* **75**: 2000–2036.

Hendry A, Nosil P, Rieseberg L. 2007. The speed of ecological speciation. *Functional Ecology* **21**: 455.

Hereford J. 2009. A quantitative survey of local adaptation and fitness trade-offs. *American Naturalist* **173**: 579–588.

Hoffmann AA, Merilä J. 1999. Heritable variation and evolution under favourable and unfavourable conditions. *Trends in Ecology & Evolution* **14**: 96–101.

Howard DJ. 1999. Conspecific sperm and pollen precedence and speciation. *Annual Review of Ecology and Systematics* **30**: 109–132.

Huberty AF, Denno RF. 2004. Plant water stress and its consequences for herbivorous insects: a new synthesis. *Ecology* **85**: 1383–1398.

Jain SK, Bradshaw AD. 1966. Evolutionary divergence among adjacent plant populations I. The evidence and its theoretical analysis. *Heredity* **21**: 407–441.

Kay KM. 2006. Reproductive isolation between two closely related hummingbird pollinated neotropical gingers. *Evolution* **60**: 538–552.

Keller I, Seehausen O. 2012. Thermal adaptation and ecological speciation. *Molecular Ecology* **21**: 782–799.

Kephart S, Theiss K. 2004. Pollinator-mediated isolation in sympatric milkweeds (*Asclepias*): do floral morphology and insect behavior influence species boundaries? *New Phytologist* **161**: 265–277.

Kirk H, Vrieling K, Van Der Meijden E, Klinkhamer P. 2010. Species by environment interactions affect pyrrolizidine alkaloid expression in *Senecio jacobaea*, *Senecio aquaticus*, and their hybrids. *Journal of Chemical Ecology* **36**: 378–387.

Kruckeberg A. 1986. An essay: the stimulus of unusual geologies for plant speciation. *Systematic Botany* **11**: 455–463.

Kruckeberg AR. 1951. Intraspecific variability in the response of certain native plant species to serpentine soil. *American Journal of Botany* **38**: 408–419.

Laland KN, Sterelny K, Odling-Smee J, Hoppitt W, Uller T. 2011. Cause and effect in biology revisited: is Mayr's proximate-ultimate dichotomy still useful? *Science* **334**: 1512–1516.

Langerhans RB, Gifford ME, Joseph EO. 2007. Ecological speciation in *Gambusia* fishes. *Evolution* **61**: 2056–2074.

Lowry D, Modliszewski J, Wright K, Wu C, Willis J. 2008a. The strength and genetic basis of reproductive isolating barriers in flowering plants. *Philosophical Transactions of the Royal Society B* **363**: 3009–3021.

Lowry DB. 2012. Ecotypes and the controversy over stages in the formation of new species. *Biological Journal of the Linnean Society* **106**: 241–257.

Lowry DB, Rockwood RC, Willis JH. 2008b. Ecological reproductive isolation of coast and inland races of *Mimulus guttatus*. *Evolution* **62**: 2196–2214.

Lowry DB, Willis J. 2010. A widespread chromosomal inversion polymorphism contributes to a major life-history transition, local adaptation, and reproductive isolation. *PLoS Biology* **8**: e100500.

Macnair M, Christie P. 1983. Reproductive isolation as a pleiotropic effect of copper tolerance in *Mimulus guttatus*. *Heredity* **50**: 295–302.

Martin NH, Bouck AC, Arnold ML. 2006. Detecting adaptive trait introgression between *Iris fulva* and *I. brevicaulis* in highly selective field conditions. *Genetics* **172**: 2481–2489.

McBride CS, Singer MC. 2010. Field studies reveal strong postmating isolation between ecologically divergent butterfly populations. *PLoS Biology* **8**: e1000529.

McNeilly T, Antonovics J. 1968. Evolution in closely adjacent plant populations IV. Barriers to gene flow. *Heredity* **23**: 205–218.

Miglia KJ, Mcarthur ED, Moore WS, Wang H, Graham JH, Freeman D. 2005. Nine-year reciprocal transplant experiment in the gardens of the basin and mountain big sagebrush (*Artemisia tridentata*: Asteraceae) hybrid zone of Salt Creek Canyon: the importance of multiple-year tracking of fitness. *Biological Journal of the Linnean Society* **86**: 213–225.

Moyle LC, Graham EB. 2005. Genetics of hybrid incompatibility between *Lycopersicon esculentum* and *L. hirsutum*. *Genetics* **169**: 355–373.

Nagy E, Rice K. 1997. Local adaptation in two subspecies of an annual plant: implications for migration and gene flow. *Evolution* **51**: 1079–1089.

Nosil P. 2004. Reproductive isolation caused by visual predation on migrants between divergent environments. *Proceedings of the Royal Society of London B* **271**: 1521–1528.

Nosil P. 2007. Divergent host plant adaptation and reproductive isolation between ecotypes of *Timema cristinae* walking sticks. *American Naturalist* **169**: 151–162.

Nosil P, Egan SP, Funk DJ. 2008. Heterogeneous genomic differentiation between walking-stick ecotypes: 'isolation by adaptation' and multiple roles for divergent selection. *Evolution* **63**: 316–336.

Nosil P, Harmon L, Seehausen O. 2009. Ecological explanations for (incomplete) speciation. *Trends in Ecology & Evolution* **24**: 145–156.

Nosil P, Vines T, Funk D. 2005. Reproductive isolation caused by natural selection against immigrants from divergent habitats. *Evolution* **59**: 705–719.

O'Dell RE, Rajakaruna N. 2011. Intraspecific variation, adaptation, and evolution. In: Harrison S, Rajakaruna N, eds. *Serpentine: the evolution and ecology of a model system*. Berkeley & Los Angeles, CA, USA: California University Press, 97–137.

Ornduff R. 1964. Evolutionary pathways of the *Senecio lautus* alliance in New Zealand and Australia. *Evolution* **18**: 349–360.

R Development Core Team. 2012. *R: a language and environment for statistical computing*. Viena, Austria: R Foundation for Statistical Computing. ISBN 3-900051-07-0, URL http://www/R-project.org.

Radford I, Cousens R, Michael P. 2004. Morphological and genetic variation in the *Senecio lautus* complex: are variants worthy of taxonomic recognition? *Australian Systematic Botany* **17**: 29–48.

Ramsey J, Bradshaw H Jr, Schemske D. 2003. Components of reproductive isolation between the monkeyflowers *Mimulus lewisii* and *M. cardinalis* (Phrymaceae). *Evolution* **57**: 1520–1534.

Roda F, Ambrose L, Walter GM, Liu H, Schaul A, Lowe A, Pelser PB, Prentis P, Rieseberg LH, Ortiz-Barrientos D. 2013a. Genomic evidence for the parallel evolution of coastal forms in the *Senecio lautus* complex. *Molecular Ecology* **22**: 2941–2952.

Roda F, Liu H, Wilkinson M, Walter G, James M, Bernal D, Melo M, Lowe A, Rieseberg L, Prentis P *et al.* 2013b. Convergence and divergence during the adaptation to similar environments by an Australian groundsel. *Evolution* **67**: 2515–2529.

Rundle HD, Nosil P. 2005. Ecological speciation. *Ecology Letters* **8**: 336–352.

Rundle HD, Whitlock MC. 2001. A genetic interpretation of ecologically dependent isolation. *Evolution* **55**: 198–201.

Sambatti J, Rice KJ. 2007. Functional ecology of ecotypic differentiation in the Californian serpentine sunflower (*Helianthus exilis*). *New Phytologist* **175**: 107–119.

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Sambatti J, Strasburg JL, Ortiz-Barrientos D, Baack EJ, Rieseberg LH. 2012. Reconciling extremly strong barriers with high levels of gene exchange in annual sunflowers. *Evolution* **66**: 1459–1473.
Schemske DW. 2010. Adaptation and the origin of species. *American Naturalist* **176**: S4–S25.
Schluter D. 1998. Ecological causes of speciation. In: Howard DJ, Stewart H, eds. *Endless forms: species and speciation*. New York, NY, USA: Oxford University Press, 114–129.
Schluter D. 2000. *Ecology of adaptive radiation*. New York, NY, USA: Oxford University Press.
Schluter D. 2001. Ecology and the origin of species. *Trends in Ecology & Evolution* **16**: 372–380.
Silvertown J, Servaes C, Biss P, Macleod D. 2005. Reinforcement of reproductive isolation between adjacent populations in the Park Grass Experiment. *Heredity* **95**: 198–205.
Sork VL, Stowe KA, Hochwender C. 1993. Evidence for local adaptation in closely adjacent subpopulations of Northern Red Oak (*Quercus rubra* L.) expressed as resistance to leaf herbivores. *American Naturalist* **142**: 928–936.
Stebbins GL. 1952. Aridity as a stimulus to plant evolution. *American Naturalist* **6**: 33–44.
Thibert-Plante X, Hendry A. 2009. Five questions on ecological speciation addressed with individual-based simulations. *Journal of Evolutionary Biology* **22**: 109–123.
Thibert-Plante X, Hendry AP. 2011. The consequences of phenotypic plasticity for ecological speciation. *Journal of Evolutionary Biology* **24**: 326–342.
Thompson I. 2005. Taxonomic studies of Australian *Senecio* (Asteraceae): 5. The *S. lautus*/*S. lautus complex*. *Muelleria* **21**: 23–76.
Vekemans X, Lefèbvre C. 1997. On the evolution of heavy-metal tolerant populations in *Armeria maritima*: evidence from allozyme variation and reproductive barriers. *Journal of Evolutionary Biology* **10**: 175–191.
Via S, Bouck A, Skillman S. 2000. Reproductive isolation between divergent races of pea aphids on two hosts. II. Selection against migrants and hybrids in the parental environments. *Evolution* **54**: 1626–1637.
Westberg E, Poppendieck H-H, Kadereit JW. 2010. Ecological differentiation and reproductive isolation of two closely related sympatric species of *Oenanthe* (Apiaceae). *Biological Journal of the Linnean Society* **101**: 526–535.
White EM. 2008. *Indirect interactions between alien and native Senecio species as mediated by insects*, PhD by Publication. Queensland University of Technology, Queensland, Australia.
Wu L, Bradshaw A, Thurman D. 1975. The potential for evolution of heavy metal tolerance in plants. III. The rapid evolution of copper tolerance in *Agrostis stolonifera*. *Heredity* **34**: 165–187.

Supporting Information

Additional supporting information may be found in the online version of this article.

Fig. S1 Typical morphologies of *Senecio lautus* populations at Cabarita beach and example of transplant experiment plot.

Fig. S2 Fruits of *Senecio lautus*.

Fig. S3 Germination in transplant experiments under controlled conditions.

Fig. S4 Examples of seedling individual in the field, a predated seedling and one of the predator individuals (*Spilosoma* sp.).

Fig. S5 Examples of *Senecio lautus* pollinators observed in both sand dune and rocky headlands field.

Table S1 Geographic location of populations used in this study

Table S2 Proportion of individuals that germinated in Dune or Headland soil under glasshouse or field conditions

Table S3 Mortality in reciprocal transplant experiments in the field

Table S4 Detailed information on statistical tests

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